$$\lambda_3 \to \frac{1}{4}(-g_1^2+g_2^2)+$$
$$\frac{1}{16\pi^2}(-\frac{1}{4}g_1^4+\frac{1}{6}g_2^4+\frac{1}{12}\sum_p g_1^4\,LF_{2,0}[m_{\tilde{d}}^{p}]-\frac{1}{2}g_1^2\,\overline{y_d}^{pr}y_d^{pr}\,LF_{2,0}[m_{\tilde{d}}^{r}]+\frac{1}{4}\sum_p g_1^4\,LF_{2,0}[m_{\tilde{e}}^{p}]-$$
$$\frac{1}{2}g_1^2\,\overline{y_e}^{pr}y_e^{pr}\,LF_{2,0}[m_{\tilde{e}}^{r}]+\frac{1}{8}(g_1^2+g_2^2)(2\,\overline{y_e}^{pr}y_e^{pr}+\sum_p(g_1^2-g_2^2))\,LF_{2,0}[m_{\tilde{l}}^{p}]+$$
$$\frac{1}{24}(-6\,\overline{y_d}^{pr}y_d^{pr}(g_1^2-3g_2^2)+6\,\overline{y_u}^{pr}y_u^{pr}(g_1^2+3g_2^2)+\sum_p(g_1^4-9g_2^4))\,LF_{2,0}[m_{\tilde{q}}^{p}]+$$
$$\frac{1}{3}\sum_p g_1^4\,LF_{2,0}[m_{\tilde{u}}^{p}]-g_1^2\,\overline{y_u}^{pr}y_u^{pr}\,LF_{2,0}[m_{\tilde{u}}^{r}]+\frac{1}{2}g_1^4\,LF_{2,2,-2}[m_1,\tilde{\mu}]+$$
$$\frac{1}{2}g_1^4m_1^2\,LF_{2,2,-1}[m_1,\tilde{\mu}]+\frac{5}{2}g_2^4\,LF_{2,2,-2}[m_2,\tilde{\mu}]+\frac{1}{2}g_2^4m_2^2\,LF_{2,2,-1}[m_2,\tilde{\mu}]+$$
$$\frac{2}{3}m_2\tilde{\mu}g_2^4(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})\,LF_{2,2,0}[m_2,\tilde{\mu}]-\frac{1}{3}m_2\tilde{\mu}g_2^4(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})\,LF_{3,2,-1}[m_2,\tilde{\mu}]+$$
$$\frac{1}{2}g_1^2(-\overline{a_d}^{pr}a_d^{pr}+\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{pr})\,LF_{2,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]+$$
$$\frac{1}{2}g_1^2(-C_{\phi2^2}\overline{a_d}^{pr}a_d^{pr}+C_{\phi1^2}\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{pr})\,LF_{2,2,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]-$$
$$3\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}\,LF_{1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}]+\frac{1}{2}g_1^2(-\overline{a_e}^{pr}a_e^{pr}+\tilde{\mu}^2\,\overline{y_e}^{pr}y_e^{pr})\,LF_{2,1,0}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]+$$
$$\frac{1}{2}g_1^2(-C_{\phi2^2}\overline{a_e}^{pr}a_e^{pr}+C_{\phi1^2}\tilde{\mu}^2\,\overline{y_e}^{pr}y_e^{pr})\,LF_{2,2,0}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]+$$
$$\frac{1}{2}(g_2^2\,\overline{a_e}^{pr}a_e^{pr}-g_1^2\tilde{\mu}^2\,\overline{y_e}^{pr}y_e^{pr})\,LF_{2,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]+\frac{1}{4}(g_1^2-g_2^2)(\overline{a_e}^{pr}a_e^{pr}+\tilde{\mu}^2\,\overline{y_e}^{pr}y_e^{pr})$$
$$LF_{3,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]+\frac{1}{2}(\tilde{\mu}\,\overline{y_e}^{pr}(\overline{C_{\phi1\phi2}}\,a_e^{pr}(-g_1^2+g_2^2)-2C_{\phi1^2}\tilde{\mu}g_1^2y_e^{pr})+$$
$$\overline{a_e}^{pr}(2C_{\phi2^2}g_2^2a_e^{pr}+C_{\phi1\phi2}\tilde{\mu}y_e^{pr}(-g_1^2+g_2^2)))\,LF_{3,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]+$$
$$\frac{1}{2}g_1^2(C_{\phi2^2}\overline{a_e}^{pr}a_e^{pr}-C_{\phi1^2}\tilde{\mu}^2\,\overline{y_e}^{pr}y_e^{pr})\,LF_{3,2,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]+$$
$$\frac{1}{4}(\overline{a_e}^{pr}(3C_{\phi2^2}a_e^{pr}(g_1^2-3g_2^2)+C_{\phi1\phi2}\tilde{\mu}y_e^{pr}(6g_1^2-5g_2^2))+$$
$$\tilde{\mu}\,\overline{y_e}^{pr}(\overline{C_{\phi1\phi2}}\,a_e^{pr}(6g_1^2-5g_2^2)+3C_{\phi1^2}\tilde{\mu}y_e^{pr}(3g_1^2-g_2^2)))\,LF_{4,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]+$$
$$\frac{1}{3}(\tilde{\mu}\,\overline{y_e}^{pr}(\overline{C_{\phi1\phi2}}\,a_e^{pr}(-3g_1^2+2g_2^2)+3C_{\phi1^2}\tilde{\mu}y_e^{pr}(-g_1^2+g_2^2))+$$
$$\overline{a_e}^{pr}(3C_{\phi2^2}a_e^{pr}(-g_1^2+g_2^2)+C_{\phi1\phi2}\tilde{\mu}y_e^{pr}(-3g_1^2+2g_2^2)))\,LF_{5,1,-2}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]+$$
$$\frac{1}{2}(\overline{a_d}^{pr}a_d^{pr}(-2g_1^2+3g_2^2)-g_1^2\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{pr})\,LF_{2,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+$$
$$\frac{3}{4}(g_1^2-g_2^2)(\overline{a_d}^{pr}a_d^{pr}+\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{pr})\,LF_{3,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+$$
$$\frac{1}{2}(\tilde{\mu}\,\overline{y_d}^{pr}(3\,\overline{C_{\phi1\phi2}}\,a_d^{pr}(-g_1^2+g_2^2)-2C_{\phi1^2}\tilde{\mu}g_1^2y_d^{pr})+$$
$$\overline{a_d}^{pr}(2C_{\phi2^2}a_d^{pr}(-2g_1^2+3g_2^2)+3C_{\phi1\phi2}\tilde{\mu}y_d^{pr}(-g_1^2+g_2^2)))\,LF_{3,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+$$
$$\frac{1}{2}g_1^2(C_{\phi2^2}\overline{a_d}^{pr}a_d^{pr}-C_{\phi1^2}\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{pr})\,LF_{3,2,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+$$
$$\frac{3}{4}(\overline{a_d}^{pr}(C_{\phi2^2}a_d^{pr}(7g_1^2-9g_2^2)+C_{\phi1\phi2}\tilde{\mu}y_d^{pr}(6g_1^2-5g_2^2))+$$
$$\tilde{\mu}\,\overline{y_d}^{pr}(\overline{C_{\phi1\phi2}}\,a_d^{pr}(6g_1^2-5g_2^2)+C_{\phi1^2}\tilde{\mu}y_d^{pr}(5g_1^2-3g_2^2)))\,LF_{4,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+$$
$$(\tilde{\mu}\,\overline{y_d}^{pr}(\overline{C_{\phi1\phi2}}\,a_d^{pr}(-3g_1^2+2g_2^2)+3C_{\phi1^2}\tilde{\mu}y_d^{pr}(-g_1^2+g_2^2))+$$
$$\overline{a_d}^{pr}(3C_{\phi2^2}a_d^{pr}(-g_1^2+g_2^2)+C_{\phi1\phi2}\tilde{\mu}y_d^{pr}(-3g_1^2+2g_2^2)))\,LF_{5,1,-2}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]-$$
$$3\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}\,LF_{1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+\frac{1}{4}(g_1^2+3g_2^2)(\overline{a_u}^{pr}a_u^{pr}-\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pr})$$
$$LF_{2,1,0}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+\frac{1}{4}(\overline{a_u}^{pr}(C_{\phi1^2}a_u^{pr}(g_1^2+3g_2^2)+2C_{\phi1\phi2}\tilde{\mu}g_2^2y_u^{pr})+$$
$$\tilde{\mu}\,\overline{y_u}^{pr}(2\,\overline{C_{\phi1\phi2}}\,g_2^2a_u^{pr}-C_{\phi2^2}\tilde{\mu}y_u^{pr}(g_1^2+3g_2^2)))\,LF_{2,2,0}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]-$$
$$\frac{1}{4}\tilde{\mu}g_2^2(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{pr}a_u^{pr}+C_{\phi1\phi2}y_u^{pr}\,\overline{a_u}^{pr})\,LF_{3,2,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+$$
$$\frac{1}{4}(\overline{a_u}^{pr}a_u^{pr}(-7g_1^2+3g_2^2)+\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pr}(g_1^2+3g_2^2))\,LF_{2,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+$$
$$\frac{3}{4}(g_1^2-g_2^2)(\overline{a_u}^{pr}a_u^{pr}+\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pr})\,LF_{3,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+$$
$$\frac{1}{2}(\overline{a_u}^{pr}(C_{\phi1^2}a_u^{pr}(-7g_1^2+3g_2^2)+C_{\phi1\phi2}\tilde{\mu}y_u^{pr}(-3g_1^2+2g_2^2))+$$
$$\tilde{\mu}\,\overline{y_u}^{pr}(\overline{C_{\phi1\phi2}}\,a_u^{pr}(-3g_1^2+2g_2^2)+C_{\phi2^2}\tilde{\mu}y_u^{pr}(g_1^2+3g_2^2)))\,LF_{3,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+$$
$$\frac{1}{4}(-\overline{a_u}^{pr}(C_{\phi1^2}a_u^{pr}(g_1^2+3g_2^2)+2C_{\phi1\phi2}\tilde{\mu}g_2^2y_u^{pr})+$$
$$\tilde{\mu}\,\overline{y_u}^{pr}(-2\,\overline{C_{\phi1\phi2}}\,g_2^2a_u^{pr}+C_{\phi2^2}\tilde{\mu}y_u^{pr}(g_1^2+3g_2^2)))\,LF_{3,2,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+$$
$$\frac{3}{2}(\tilde{\mu}\,\overline{y_u}^{pr}(\overline{C_{\phi1\phi2}}\,a_u^{pr}(3g_1^2-2g_2^2)+C_{\phi2^2}\tilde{\mu}y_u^{pr}(g_1^2-3g_2^2))+$$
$$\overline{a_u}^{pr}(C_{\phi1^2}a_u^{pr}(5g_1^2-3g_2^2)+C_{\phi1\phi2}\tilde{\mu}y_u^{pr}(3g_1^2-2g_2^2)))\,LF_{4,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+$$
$$(\tilde{\mu}\,\overline{y_u}^{pr}(\overline{C_{\phi1\phi2}}\,a_u^{pr}(-3g_1^2+2g_2^2)+3C_{\phi2^2}\tilde{\mu}y_u^{pr}(-g_1^2+g_2^2))+$$
$$\overline{a_u}^{pr}(3C_{\phi1^2}a_u^{pr}(-g_1^2+g_2^2)+C_{\phi1\phi2}\tilde{\mu}y_u^{pr}(-3g_1^2+2g_2^2)))\,LF_{5,1,-2}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]-$$
$$3\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(C_{\phi1^2}+C_{\phi2^2})\,LF_{2,1,0}[m_{\tilde{u}}^{t},m_{\tilde{d}}^{r}]+3\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}$$
$$(C_{\phi1^2}+C_{\phi2^2})\,LF_{3,1,-1}[m_{\tilde{u}}^{t},m_{\tilde{d}}^{r}]+\frac{3}{4}(g_1^4-g_1^2g_2^2)\,LF_{2,1,-1}[\tilde{\mu},m_1]+$$
$$\frac{1}{2}(-g_1^4+g_1^2g_2^2)\,LF_{3,1,-2}[\tilde{\mu},m_1]+g_1^2(C_{\phi1^2}+C_{\phi2^2})(g_1^2-g_2^2)\,LF_{3,1,-1}[\tilde{\mu},m_1]-$$
$$\frac{1}{2}m_1\tilde{\mu}g_1^2(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(g_1^2-g_2^2)\,LF_{3,1,0}[\tilde{\mu},m_1]+\frac{3}{4}g_1^4(C_{\phi1^2}+C_{\phi2^2})\,LF_{3,2,-2}[\tilde{\mu},m_1]+$$
$$\frac{1}{4}m_1g_1^4(3m_1(C_{\phi1^2}+C_{\phi2^2})-4\tilde{\mu}(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2}))\,LF_{3,2,-1}[\tilde{\mu},m_1]-$$
$$2g_1^2(C_{\phi1^2}+C_{\phi2^2})(g_1^2-g_2^2)\,LF_{4,1,-2}[\tilde{\mu},m_1]+$$
$$\frac{1}{4}m_1\tilde{\mu}g_1^2(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(6g_1^2-5g_2^2)\,LF_{4,1,-1}[\tilde{\mu},m_1]-\frac{1}{2}g_1^4(C_{\phi1^2}+C_{\phi2^2})\,LF_{4,2,-3}[\tilde{\mu},m_1]+$$
$$\frac{1}{2}m_1g_1^4(-m_1(C_{\phi1^2}+C_{\phi2^2})+2\tilde{\mu}(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2}))\,LF_{4,2,-2}[\tilde{\mu},m_1]+$$
$$g_1^2(C_{\phi1^2}+C_{\phi2^2})(g_1^2-g_2^2)\,LF_{5,1,-3}[\tilde{\mu},m_1]-$$
$$\frac{1}{3}m_1\tilde{\mu}g_1^2(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(3g_1^2-2g_2^2)\,LF_{5,1,-2}[\tilde{\mu},m_1]+\frac{9}{4}g_2^2(g_1^2-g_2^2)\,LF_{2,1,-1}[\tilde{\mu},m_2]+$$
$$\frac{3}{2}g_2^2(-g_1^2+g_2^2)\,LF_{3,1,-2}[\tilde{\mu},m_2]+3g_2^2(C_{\phi1^2}+C_{\phi2^2})(g_1^2-g_2^2)\,LF_{3,1,-1}[\tilde{\mu},m_2]+$$
$$\frac{1}{6}m_2\tilde{\mu}g_2^2(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(-9g_1^2+5g_2^2)\,LF_{3,1,0}[\tilde{\mu},m_2]+\frac{15}{4}g_2^4(C_{\phi1^2}+C_{\phi2^2})\,LF_{3,2,-2}[\tilde{\mu},m_2]+$$
$$\frac{1}{12}m_2g_2^4(9m_2(C_{\phi1^2}+C_{\phi2^2})-56\tilde{\mu}(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2}))\,LF_{3,2,-1}[\tilde{\mu},m_2]+$$
$$6g_2^2(C_{\phi1^2}+C_{\phi2^2})(-g_1^2+g_2^2)\,LF_{4,1,-2}[\tilde{\mu},m_2]+$$
$$\frac{1}{4}m_2\tilde{\mu}g_2^2(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(18g_1^2-11g_2^2)\,LF_{4,1,-1}[\tilde{\mu},m_2]-\frac{5}{2}g_2^4(C_{\phi1^2}+C_{\phi2^2})\,LF_{4,2,-3}[\tilde{\mu},m_2]+$$
$$\frac{1}{2}g_2^4(-m_2^2(C_{\phi1^2}+C_{\phi2^2})+8m_2\tilde{\mu}(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})-2\tilde{\mu}^2(C_{\phi1^2}+C_{\phi2^2}))\,LF_{4,2,-2}[\tilde{\mu},m_2]+$$
$$3g_2^2(C_{\phi1^2}+C_{\phi2^2})(g_1^2-g_2^2)\,LF_{5,1,-3}[\tilde{\mu},m_2]+$$
$$m_2\tilde{\mu}g_2^2(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(-3g_1^2+2g_2^2)\,LF_{5,1,-2}[\tilde{\mu},m_2]+$$
$$\frac{3}{2}\tilde{\mu}g_1^2g_2^2(-(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(3m_1+m_2)+4\tilde{\mu}(C_{\phi1^2}+C_{\phi2^2}))\,LF_{2,2,1,-1}[m_2,\tilde{\mu},m_1]+$$
$$2\tilde{\mu}g_1^2g_2^2((\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(m_1+m_2)-2\tilde{\mu}(C_{\phi1^2}+C_{\phi2^2}))\,LF_{3,2,1,-2}[m_2,\tilde{\mu},m_1]-$$
$$3\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,LF_{1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}]+$$
$$\frac{3}{4}\tilde{\mu}\,\overline{y_d}^{st}y_d^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr}a_d^{pt}-C_{\phi1\phi2}y_d^{pt}\,\overline{a_d}^{pr})\,LF_{2,2,1,-1}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{d}}^{t}]-$$
$$3\,\overline{y_u}^{st}y_u^{pt}\,\overline{a_d}^{pr}a_d^{sr}\,LF_{1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+$$
$$3y_d^{sr}\,\overline{y_u}^{st}(-\overline{a_d}^{pr}a_u^{pt}+\tilde{\mu}^2\,\overline{y_d}^{pr}y_u^{pt})\,LF_{1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]+$$
$$3\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}\,LF_{1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$
$$\frac{3}{4}\tilde{\mu}\,\overline{y_d}^{st}y_d^{sr}(-\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr}a_d^{pt}+C_{\phi1\phi2}y_d^{pt}\,\overline{a_d}^{pr})\,LF_{2,2,1,-1}[m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]-$$
$$3\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}\,LF_{1,1,1,0}[m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]-\tilde{\mu}^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,LF_{1,1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{s}]+$$
$$\frac{1}{4}\tilde{\mu}\,\overline{y_e}^{st}y_e^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_e}^{pr}a_e^{pt}-C_{\phi1\phi2}y_e^{pt}\,\overline{a_e}^{pr})\,LF_{2,2,1,-1}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p},m_{\tilde{e}}^{t}]+$$
$$\frac{1}{4}\tilde{\mu}\,\overline{y_e}^{st}y_e^{sr}(-\overline{C_{\phi1\phi2}}\,\overline{y_e}^{pr}a_e^{pt}+C_{\phi1\phi2}y_e^{pt}\,\overline{a_e}^{pr})\,LF_{2,2,1,-1}[m_{\tilde{e}}^{t},m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]-$$
$$\frac{1}{2}\tilde{\mu}\,\overline{y_e}^{st}y_e^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_e}^{pr}a_e^{pt}+C_{\phi1\phi2}y_e^{pt}\,\overline{a_e}^{pr})\,LF_{2,1,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t}]+$$
$$\frac{1}{2}\tilde{\mu}\,\overline{y_e}^{st}y_e^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_e}^{pr}a_e^{pt}+C_{\phi1\phi2}y_e^{pt}\,\overline{a_e}^{pr})\,LF_{3,1,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t}]-$$
$$\frac{1}{4}C_{\phi1\phi2}\tilde{\mu}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\overline{a_e}^{pr}\,LF_{2,2,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{l}}^{s},m_{\tilde{e}}^{r}]-$$
$$\frac{1}{4}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{sr}a_e^{pt}\,LF_{2,2,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{l}}^{s},m_{\tilde{e}}^{t}]-$$
$$C_{\phi1^2}\tilde{\mu}^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,LF_{2,1,1,0}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t}]+$$
$$C_{\phi1^2}\tilde{\mu}^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,LF_{3,1,1,-1}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t}]-$$
$$\frac{3}{2}\tilde{\mu}\,\overline{y_d}^{st}y_d^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr}a_d^{pt}+C_{\phi1\phi2}y_d^{pt}\,\overline{a_d}^{pr})\,LF_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t}]+$$
$$\frac{3}{2}\tilde{\mu}\,\overline{y_d}^{st}y_d^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr}a_d^{pt}+C_{\phi1\phi2}y_d^{pt}\,\overline{a_d}^{pr})\,LF_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t}]-$$
$$3\,\overline{y_u}^{st}y_u^{pt}a_d^{sr}(C_{\phi2^2}\overline{a_d}^{pr}+\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_d}^{pr})\,LF_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}]+$$
$$3\,\overline{y_u}^{st}y_u^{pt}a_d^{sr}(C_{\phi2^2}\overline{a_d}^{pr}+\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_d}^{pr})\,LF_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}]+3y_d^{sr}\,\overline{y_u}^{st}$$
$$(\tilde{\mu}\,\overline{y_d}^{pr}(-\overline{C_{\phi1\phi2}}\,a_u^{pt}+C_{\phi1^2}\tilde{\mu}y_u^{pt})+\overline{a_d}^{pr}(-C_{\phi2^2}a_u^{pt}+C_{\phi1\phi2}\tilde{\mu}y_u^{pt}))\,LF_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}]+$$
$$3y_d^{sr}\,\overline{y_u}^{st}(\tilde{\mu}\,\overline{y_d}^{pr}(\overline{C_{\phi1\phi2}}\,a_u^{pt}-C_{\phi1^2}\tilde{\mu}y_u^{pt})+\overline{a_d}^{pr}(C_{\phi2^2}a_u^{pt}-C_{\phi1\phi2}\tilde{\mu}y_u^{pt}))$$
$$LF_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}]-3\,\overline{y_d}^{st}y_u^{sr}\,\overline{a_u}^{pr}(C_{\phi2^2}a_d^{pt}+C_{\phi1\phi2}\tilde{\mu}y_d^{pt})\,LF_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{u}}^{r}]+$$
$$3\,\overline{y_d}^{st}y_u^{sr}\,\overline{a_u}^{pr}(C_{\phi2^2}a_d^{pt}+C_{\phi1\phi2}\tilde{\mu}y_d^{pt})\,LF_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{u}}^{r}]+$$
$$\frac{3}{4}\overline{a_d}^{pr}(4C_{\phi2^2}\overline{y_u}^{st}y_u^{pt}a_d^{sr}-C_{\phi1\phi2}\tilde{\mu}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr})\,LF_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{r}]-$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{sr}a_d^{pt}\,LF_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{t}]-$$
$$3\,\overline{y_d}^{st}y_d^{pt}\,\overline{a_u}^{pr}a_u^{sr}\,LF_{1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+$$
$$\frac{3}{4}C_{\phi1\phi2}\tilde{\mu}y_u^{sr}\,\overline{a_u}^{pr}(\overline{y_d}^{st}y_d^{pt}-\overline{y_u}^{st}y_u^{pt})\,LF_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+$$
$$\frac{3}{4}C_{\phi1\phi2}\tilde{\mu}y_u^{sr}\,\overline{a_u}^{pr}(-\overline{y_d}^{st}y_d^{pt}+\overline{y_u}^{st}y_u^{pt})\,LF_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{st}a_u^{pt}(\overline{y_d}^{pr}y_d^{sr}-\overline{y_u}^{pr}y_u^{sr})\,LF_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{st}a_u^{pt}(-\overline{y_d}^{pr}y_d^{sr}+\overline{y_u}^{pr}y_u^{sr})\,LF_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$
$$3\,\overline{y_d}^{st}y_u^{sr}\,\overline{a_u}^{pr}(C_{\phi2^2}a_d^{pt}+C_{\phi1\phi2}\tilde{\mu}y_d^{pt})\,LF_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{d}}^{t}]+$$
$$\frac{3}{8}C_{\phi1\phi2}\tilde{\mu}y_u^{sr}\,\overline{a_u}^{pr}(\overline{y_d}^{st}y_d^{pt}+\overline{y_u}^{st}y_u^{pt})\,LF_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{q}}^{s}]+3y_d^{sr}\,\overline{y_u}^{st}$$
$$(\tilde{\mu}\,\overline{y_d}^{pr}(\overline{C_{\phi1\phi2}}\,a_u^{pt}-C_{\phi1^2}\tilde{\mu}y_u^{pt})+\overline{a_d}^{pr}(C_{\phi2^2}a_u^{pt}-C_{\phi1\phi2}\tilde{\mu}y_u^{pt}))\,LF_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{d}}^{r}]+$$
$$\frac{3}{8}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{st}a_u^{pt}(\overline{y_d}^{pr}y_d^{sr}+\overline{y_u}^{pr}y_u^{sr})\,LF_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{q}}^{s}]-$$
$$3C_{\phi1^2}\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,LF_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t}]+$$
$$3C_{\phi1^2}\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,LF_{3,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t}]-$$
$$3\,\overline{y_u}^{st}y_u^{pt}\,\overline{a_d}^{pr}(C_{\phi2^2}a_d^{sr}+C_{\phi1\phi2}\tilde{\mu}y_d^{sr})\,LF_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]+$$
$$3\,\overline{y_u}^{st}y_u^{pt}\,\overline{a_d}^{pr}(C_{\phi2^2}a_d^{sr}+C_{\phi1\phi2}\tilde{\mu}y_d^{sr})\,LF_{3,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]+$$
$$3\tilde{\mu}\,\overline{y_d}^{pr}\,\overline{y_u}^{st}y_u^{pt}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde{\mu}y_d^{sr})\,LF_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}]-$$
$$3\tilde{\mu}\,\overline{y_d}^{pr}\,\overline{y_u}^{st}y_u^{pt}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde{\mu}y_d^{sr})\,LF_{3,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}]+$$
$$\frac{9}{4}\tilde{\mu}(\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr}\,\overline{y_u}^{st}y_u^{pt}a_d^{sr}+C_{\phi1\phi2}y_d^{pr}\,\overline{y_u}^{pt}y_u^{st}\,\overline{a_d}^{sr})\,LF_{2,2,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+$$
$$\frac{3}{4}C_{\phi1\phi2}\tilde{\mu}y_u^{sr}\,\overline{a_u}^{pr}(\overline{y_d}^{st}y_d^{pt}-\overline{y_u}^{st}y_u^{pt})\,LF_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+$$
$$\frac{3}{4}C_{\phi1\phi2}\tilde{\mu}y_u^{sr}\,\overline{a_u}^{pr}(-\overline{y_d}^{st}y_d^{pt}+\overline{y_u}^{st}y_u^{pt})\,LF_{3,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{st}a_u^{pt}(\overline{y_d}^{pr}y_d^{sr}-\overline{y_u}^{pr}y_u^{sr})\,LF_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]+$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{st}a_u^{pt}(-\overline{y_d}^{pr}y_d^{sr}+\overline{y_u}^{pr}y_u^{sr})\,LF_{3,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]+$$
$$\frac{3}{8}C_{\phi1\phi2}\tilde{\mu}y_u^{sr}\,\overline{a_u}^{pr}(-3\,\overline{y_d}^{st}y_d^{pt}+\overline{y_u}^{st}y_u^{pt})\,LF_{2,2,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]-$$
$$3\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,LF_{1,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]-$$
$$3\tilde{\mu}\,\overline{y_d}^{pr}\,\overline{y_u}^{st}y_u^{pt}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde{\mu}y_d^{sr})\,LF_{2,2,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{d}}^{r}]+$$
$$\frac{3}{8}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{st}a_u^{pt}(-3\,\overline{y_d}^{pr}y_d^{sr}+\overline{y_u}^{pr}y_u^{sr})\,LF_{2,2,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p}]-$$
$$3\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(C_{\phi1^2}+C_{\phi2^2})\,LF_{2,1,1,0}[m_{\tilde{u}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+$$
$$3\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(C_{\phi1^2}+C_{\phi2^2})\,LF_{3,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]-$$
$$\frac{3}{2}\overline{y_d}^{st}y_d^{pt}\,\overline{a_u}^{pr}(2C_{\phi1^2}a_u^{sr}+C_{\phi1\phi2}\tilde{\mu}y_u^{sr})\,LF_{2,1,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+$$
$$\frac{3}{2}\overline{y_d}^{st}y_d^{pt}\,\overline{a_u}^{pr}(2C_{\phi1^2}a_u^{sr}+C_{\phi1\phi2}\tilde{\mu}y_u^{sr})\,LF_{3,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]-$$
$$3C_{\phi1\phi2}\tilde{\mu}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,\overline{a_u}^{pr}\,LF_{2,1,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]+$$
$$3C_{\phi1\phi2}\tilde{\mu}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,\overline{a_u}^{pr}\,LF_{3,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]-$$
$$3C_{\phi2^2}\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,LF_{2,1,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$
$$3C_{\phi2^2}\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,LF_{3,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$
$$\frac{3}{2}\tilde{\mu}\,\overline{y_u}^{st}y_u^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{pr}a_u^{pt}+C_{\phi1\phi2}y_u^{pt}\,\overline{a_u}^{pr})\,LF_{2,2,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p}]+$$
$$3C_{\phi2^2}\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,LF_{2,2,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{u}}^{t},m_{\tilde{q}}^{s}]-$$
$$3y_d^{sr}\,\overline{y_u}^{st}(C_{\phi1^2}+C_{\phi2^2})(\overline{a_d}^{pr}a_u^{pt}-\tilde{\mu}^2\,\overline{y_d}^{pr}y_u^{pt})\,LF_{2,1,1,0}[m_{\tilde{u}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]+$$
$$3y_d^{sr}\,\overline{y_u}^{st}(C_{\phi1^2}+C_{\phi2^2})(\overline{a_d}^{pr}a_u^{pt}-\tilde{\mu}^2\,\overline{y_d}^{pr}y_u^{pt})\,LF_{3,1,1,-1}[m_{\tilde{u}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]+$$
$$3\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(C_{\phi1^2}+C_{\phi2^2})\,LF_{2,1,1,0}[m_{\tilde{u}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}]-$$
$$3\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(C_{\phi1^2}+C_{\phi2^2})\,LF_{3,1,1,-1}[m_{\tilde{u}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}]-$$
$$\frac{3}{2}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}a_u^{pt}\,LF_{2,1,1,0}[m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+$$
$$\frac{3}{2}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}a_u^{pt}\,LF_{3,1,1,-1}[m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]-$$
$$3\,\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}\,LF_{2,1,1,0}[m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+$$
$$3\,\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}\,LF_{3,1,1,-1}[m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]-$$
$$3C_{\phi2^2}\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,LF_{2,1,1,0}[m_{\tilde{u}}^{t},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+$$
$$3C_{\phi2^2}\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,LF_{3,1,1,-1}[m_{\tilde{u}}^{t},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]-$$
$$g_1^2g_2^2\,LF_{2,1,1,-2}[\tilde{\mu},m_1,m_2]+g_1^2g_2^2(-m_1m_2+4\tilde{\mu}^2)\,LF_{2,1,1,-1}[\tilde{\mu},m_1,m_2]+$$
$$3m_1\tilde{\mu}g_1^2g_2^2(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})\,LF_{2,1,1,0}[\tilde{\mu},m_1,m_2]-\frac{3}{2}g_1^2g_2^2(C_{\phi1^2}+C_{\phi2^2})\,LF_{3,1,1,-2}[\tilde{\mu},m_1,m_2]+$$
$$\frac{1}{4}g_1^2g_2^2(-6m_1m_2(C_{\phi1^2}+C_{\phi2^2})-\tilde{\mu}(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(31m_1-5m_2)+24\tilde{\mu}^2(C_{\phi1^2}+C_{\phi2^2}))$$
$$LF_{3,1,1,-1}[\tilde{\mu},m_1,m_2]+g_1^2g_2^2(C_{\phi1^2}+C_{\phi2^2})\,LF_{4,1,1,-3}[\tilde{\mu},m_1,m_2]+$$
$$\frac{1}{2}g_1^2g_2^2(2m_1m_2(C_{\phi1^2}+C_{\phi2^2})+3\tilde{\mu}(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(3m_1-m_2)-10\tilde{\mu}^2(C_{\phi1^2}+C_{\phi2^2}))$$
$$LF_{4,1,1,-2}[\tilde{\mu},m_1,m_2]+4\tilde{\mu}g_1^2g_2^2(m_1(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})-\tilde{\mu}(C_{\phi1^2}+C_{\phi2^2}))\,LF_{3,2,1,-2}[\tilde{\mu},m_2,m_1]-$$
$$3\tilde{\mu}^2\,\overline{y_d}^{st}y_d^{sr}\,\overline{a_d}^{pr}a_d^{pt}\,LF_{1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+$$
$$\frac{3}{4}\tilde{\mu}^3\,\overline{y_d}^{st}y_d^{sr}(-\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr}a_d^{pt}+C_{\phi1\phi2}y_d^{pt}\,\overline{a_d}^{pr})\,LF_{2,2,1,1,-1}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}]-$$
$$3(\tilde{\mu}^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}+\overline{a_d}^{pr}a_u^{pt}(a_d^{sr}\,\overline{a_u}^{st}-\tilde{\mu}^2y_d^{sr}\,\overline{y_u}^{st}))\,LF_{1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$
$$\frac{3}{4}\tilde{\mu}\,\overline{a_d}^{pr}a_d^{pt}(-\overline{C_{\phi1\phi2}}\,\overline{y_d}^{st}a_d^{sr}+C_{\phi1\phi2}y_d^{sr}\,\overline{a_d}^{st})\,LF_{2,2,1,1,-1}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+$$
$$\frac{3}{4}\tilde{\mu}^3\,\overline{y_d}^{st}y_d^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr}a_d^{pt}-C_{\phi1\phi2}y_d^{pt}\,\overline{a_d}^{pr})\,LF_{2,2,1,1,-1}[m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}]+$$
$$3\tilde{\mu}^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}\,LF_{1,1,1,1,0}[m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+$$
$$\frac{3}{4}\tilde{\mu}\,\overline{a_d}^{pr}a_d^{pt}(\overline{C_{\phi1\phi2}}\,\overline{y_d}^{st}a_d^{sr}-C_{\phi1\phi2}y_d^{sr}\,\overline{a_d}^{st})\,LF_{2,2,1,1,-1}[m_{\tilde{d}}^{t},m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]-$$
$$\tilde{\mu}^2\,\overline{y_e}^{st}y_e^{sr}\,\overline{a_e}^{pr}a_e^{pt}\,LF_{1,1,1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}]+$$
$$\frac{1}{4}\tilde{\mu}^3\,\overline{y_e}^{st}y_e^{sr}(-\overline{C_{\phi1\phi2}}\,\overline{y_e}^{pr}a_e^{pt}+C_{\phi1\phi2}y_e^{pt}\,\overline{a_e}^{pr})\,LF_{2,2,1,1,-1}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p},m_{\tilde{e}}^{t},m_{\tilde{l}}^{s}]+$$
$$\frac{1}{4}\tilde{\mu}\,\overline{a_e}^{pr}a_e^{pt}(-\overline{C_{\phi1\phi2}}\,\overline{y_e}^{st}a_e^{sr}+C_{\phi1\phi2}y_e^{sr}\,\overline{a_e}^{st})\,LF_{2,2,1,1,-1}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{s},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]+$$
$$\frac{1}{4}\tilde{\mu}^3\,\overline{y_e}^{st}y_e^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_e}^{pr}a_e^{pt}-C_{\phi1\phi2}y_e^{pt}\,\overline{a_e}^{pr})\,LF_{2,2,1,1,-1}[m_{\tilde{e}}^{t},m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{l}}^{s}]+$$
$$\frac{1}{4}\tilde{\mu}\,\overline{a_e}^{pr}a_e^{pt}(\overline{C_{\phi1\phi2}}\,\overline{y_e}^{st}a_e^{sr}-C_{\phi1\phi2}y_e^{sr}\,\overline{a_e}^{st})\,LF_{2,2,1,1,-1}[m_{\tilde{e}}^{t},m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]-\frac{1}{2}\tilde{\mu}^2\,\overline{y_e}^{st}y_e^{sr}$$
$$(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_e}^{pr}a_e^{pt}+\overline{a_e}^{pr}(2C_{\phi2^2}a_e^{pt}+C_{\phi1\phi2}\tilde{\mu}y_e^{pt}))\,LF_{2,1,1,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{s}]+\frac{1}{2}\tilde{\mu}^2\,\overline{y_e}^{st}$$
$$y_e^{sr}(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_e}^{pr}a_e^{pt}+\overline{a_e}^{pr}(2C_{\phi2^2}a_e^{pt}+C_{\phi1\phi2}\tilde{\mu}y_e^{pt}))\,LF_{3,1,1,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{s}]-$$
$$\frac{1}{2}\tilde{\mu}\,\overline{a_e}^{pr}a_e^{pt}(C_{\phi1\phi2}y_e^{sr}\,\overline{a_e}^{st}+\overline{y_e}^{st}(\overline{C_{\phi1\phi2}}\,a_e^{sr}+2C_{\phi1^2}\tilde{\mu}y_e^{sr}))\,LF_{2,1,1,1,0}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]+$$
$$\frac{1}{2}\tilde{\mu}\,\overline{a_e}^{pr}a_e^{pt}(C_{\phi1\phi2}y_e^{sr}\,\overline{a_e}^{st}+\overline{y_e}^{st}(\overline{C_{\phi1\phi2}}\,a_e^{sr}+2C_{\phi1^2}\tilde{\mu}y_e^{sr}))$$
$$LF_{3,1,1,1,-1}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]-\frac{3}{2}\tilde{\mu}^2\,\overline{y_d}^{st}y_d^{sr}$$
$$(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_d}^{pr}a_d^{pt}+\overline{a_d}^{pr}(2C_{\phi2^2}a_d^{pt}+C_{\phi1\phi2}\tilde{\mu}y_d^{pt}))\,LF_{2,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}]+$$
$$\frac{3}{2}\tilde{\mu}^2\,\overline{y_d}^{st}y_d^{sr}(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_d}^{pr}a_d^{pt}+\overline{a_d}^{pr}(2C_{\phi2^2}a_d^{pt}+C_{\phi1\phi2}\tilde{\mu}y_d^{pt}))$$
$$LF_{3,1,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}]+3(\tilde{\mu}^3\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}(\overline{C_{\phi1\phi2}}\,a_u^{pt}-C_{\phi1^2}\tilde{\mu}y_u^{pt})+$$
$$\overline{a_d}^{pr}(C_{\phi2^2}a_u^{pt}-C_{\phi1\phi2}\tilde{\mu}y_u^{pt})(-a_d^{sr}\,\overline{a_u}^{st}+\tilde{\mu}^2y_d^{sr}\,\overline{y_u}^{st}))\,LF_{2,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$
$$3(\tilde{\mu}^3\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}(-\overline{C_{\phi1\phi2}}\,a_u^{pt}+C_{\phi1^2}\tilde{\mu}y_u^{pt})+\overline{a_d}^{pr}(C_{\phi2^2}a_u^{pt}-C_{\phi1\phi2}\tilde{\mu}y_u^{pt})$$
$$(a_d^{sr}\,\overline{a_u}^{st}-\tilde{\mu}^2y_d^{sr}\,\overline{y_u}^{st}))\,LF_{3,1,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$
$$3\tilde{\mu}^2\,\overline{y_d}^{st}y_u^{sr}\,\overline{a_u}^{pr}(C_{\phi2^2}a_d^{pt}+C_{\phi1\phi2}\tilde{\mu}y_d^{pt})\,LF_{2,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]-$$
$$3\tilde{\mu}^2\,\overline{y_d}^{st}y_u^{sr}\,\overline{a_u}^{pr}(C_{\phi2^2}a_d^{pt}+C_{\phi1\phi2}\tilde{\mu}y_d^{pt})\,LF_{3,1,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]-$$
$$\frac{3}{4}(C_{\phi1\phi2}\tilde{\mu}y_u^{st}\,\overline{a_d}^{sr}a_d^{pr}\,\overline{a_u}^{pt}+\tilde{\mu}^3\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}(\overline{C_{\phi1\phi2}}\,a_u^{pt}-4C_{\phi1^2}\tilde{\mu}y_u^{pt})+\overline{a_d}^{pr}a_u^{pt}$$
$$(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{st}a_d^{sr}-\overline{a_u}^{st}(4C_{\phi2^2}a_d^{sr}+3C_{\phi1\phi2}\tilde{\mu}y_d^{sr})))\,LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}]-$$
$$\frac{3}{4}\tilde{\mu}\,\overline{y_d}^{st}\,\overline{a_u}^{pr}(-3\,\overline{C_{\phi1\phi2}}\,a_d^{pt}a_u^{sr}+C_{\phi1\phi2}\tilde{\mu}^2y_d^{pt}y_u^{sr})\,LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{u}}^{r}]-$$
$$3\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{sr}\,\overline{a_u}^{pr}a_u^{pt}\,LF_{1,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]-$$
$$\frac{3}{4}\tilde{\mu}y_u^{sr}\,\overline{a_u}^{pr}(C_{\phi1\phi2}\overline{a_u}^{st}a_u^{pt}+\tilde{\mu}\,\overline{y_u}^{st}(2a_u^{pt}(C_{\phi1^2}+C_{\phi2^2})+C_{\phi1\phi2}\tilde{\mu}y_u^{pt}))$$
$$LF_{2,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]+\frac{3}{4}\tilde{\mu}y_u^{sr}\,\overline{a_u}^{pr}$$
$$(C_{\phi1\phi2}\overline{a_u}^{st}a_u^{pt}+\tilde{\mu}\,\overline{y_u}^{st}(2a_u^{pt}(C_{\phi1^2}+C_{\phi2^2})+C_{\phi1\phi2}\tilde{\mu}y_u^{pt}))\,LF_{3,1,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]-$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{st}a_u^{pt}(\overline{a_u}^{pr}a_u^{sr}+\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{sr})\,LF_{2,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}]+$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{st}a_u^{pt}(\overline{a_u}^{pr}a_u^{sr}+\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{sr})\,LF_{3,1,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}]-$$
$$3\tilde{\mu}^2\,\overline{y_d}^{st}y_u^{sr}\,\overline{a_u}^{pr}(C_{\phi2^2}a_d^{pt}+C_{\phi1\phi2}\tilde{\mu}y_d^{pt})\,LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}]+$$
$$\frac{3}{8}(-\overline{C_{\phi1\phi2}}\,\tilde{\mu}^3\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}+$$
$$\tilde{\mu}\,\overline{a_u}^{pr}(3\,\overline{C_{\phi1\phi2}}\,\overline{y_u}^{st}a_u^{pt}a_u^{sr}+y_u^{sr}(C_{\phi1\phi2}\overline{a_u}^{st}a_u^{pt}+\tilde{\mu}\,\overline{y_u}^{st}(4C_{\phi2^2}a_u^{pt}+C_{\phi1\phi2}\tilde{\mu}y_u^{pt}))))$$
$$LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+3(\tilde{\mu}^3\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}(-\overline{C_{\phi1\phi2}}\,a_u^{pt}+C_{\phi1^2}\tilde{\mu}y_u^{pt})+$$
$$\overline{a_d}^{pr}(C_{\phi2^2}a_u^{pt}-C_{\phi1\phi2}\tilde{\mu}y_u^{pt})(a_d^{sr}\,\overline{a_u}^{st}-\tilde{\mu}^2y_d^{sr}\,\overline{y_u}^{st}))$$
$$LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}]+\frac{3}{8}(\overline{C_{\phi1\phi2}}\,\tilde{\mu}^3\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}+$$
$$\tilde{\mu}\,\overline{a_u}^{pr}(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{st}a_u^{pt}a_u^{sr}+y_u^{sr}(3C_{\phi1\phi2}\overline{a_u}^{st}a_u^{pt}+\tilde{\mu}\,\overline{y_u}^{st}(4C_{\phi2^2}a_u^{pt}-C_{\phi1\phi2}\tilde{\mu}y_u^{pt}))))$$
$$LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]-\frac{3}{2}\tilde{\mu}\,\overline{a_d}^{pr}a_d^{pt}$$
$$(C_{\phi1\phi2}y_d^{sr}\,\overline{a_d}^{st}+\overline{y_d}^{st}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+2C_{\phi1^2}\tilde{\mu}y_d^{sr}))\,LF_{2,1,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+$$
$$\frac{3}{2}\tilde{\mu}\,\overline{a_d}^{pr}a_d^{pt}(C_{\phi1\phi2}y_d^{sr}\,\overline{a_d}^{st}+\overline{y_d}^{st}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+2C_{\phi1^2}\tilde{\mu}y_d^{sr}))$$
$$LF_{3,1,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]-3(\tilde{\mu}^3\,\overline{y_d}^{pr}\,\overline{y_u}^{st}y_u^{pt}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde{\mu}y_d^{sr})+$$
$$\overline{a_d}^{pr}a_u^{pt}(-\tilde{\mu}\,\overline{y_u}^{st}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde{\mu}y_d^{sr})+\overline{a_u}^{st}(C_{\phi2^2}a_d^{sr}+C_{\phi1\phi2}\tilde{\mu}y_d^{sr})))$$
$$LF_{2,1,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]+3(\tilde{\mu}^3\,\overline{y_d}^{pr}\,\overline{y_u}^{st}y_u^{pt}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde{\mu}y_d^{sr})+$$
$$\overline{a_d}^{pr}a_u^{pt}(-\tilde{\mu}\,\overline{y_u}^{st}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde{\mu}y_d^{sr})+\overline{a_u}^{st}(C_{\phi2^2}a_d^{sr}+C_{\phi1\phi2}\tilde{\mu}y_d^{sr})))$$
$$LF_{3,1,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]+3\tilde{\mu}\,\overline{y_d}^{st}a_d^{pt}\,\overline{a_u}^{pr}(-\overline{C_{\phi1\phi2}}\,a_u^{sr}+C_{\phi1^2}\tilde{\mu}y_u^{sr})$$
$$LF_{2,1,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+$$
$$3\tilde{\mu}\,\overline{y_d}^{st}a_d^{pt}\,\overline{a_u}^{pr}(\overline{C_{\phi1\phi2}}\,a_u^{sr}-C_{\phi1^2}\tilde{\mu}y_u^{sr})\,LF_{3,1,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+$$
$$\frac{9}{4}\tilde{\mu}^3(\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr}\,\overline{y_u}^{st}y_u^{pt}a_d^{sr}+C_{\phi1\phi2}y_d^{pr}\,\overline{y_u}^{pt}y_u^{st}\,\overline{a_d}^{sr})\,LF_{2,2,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}]-$$
$$\frac{3}{4}\tilde{\mu}y_u^{sr}\,\overline{a_u}^{pr}(C_{\phi1\phi2}\overline{a_u}^{st}a_u^{pt}+\tilde{\mu}\,\overline{y_u}^{st}(2a_u^{pt}(C_{\phi1^2}+C_{\phi2^2})+C_{\phi1\phi2}\tilde{\mu}y_u^{pt}))$$
$$LF_{2,1,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]+\frac{3}{4}\tilde{\mu}y_u^{sr}\,\overline{a_u}^{pr}$$
$$(C_{\phi1\phi2}\overline{a_u}^{st}a_u^{pt}+\tilde{\mu}\,\overline{y_u}^{st}(2a_u^{pt}(C_{\phi1^2}+C_{\phi2^2})+C_{\phi1\phi2}\tilde{\mu}y_u^{pt}))\,LF_{3,1,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]-$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{st}a_u^{pt}(\overline{a_u}^{pr}a_u^{sr}+\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{sr})\,LF_{2,1,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}]+$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{st}a_u^{pt}(\overline{a_u}^{pr}a_u^{sr}+\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{sr})\,LF_{3,1,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}]+$$
$$3\tilde{\mu}\,\overline{y_d}^{st}a_d^{pt}\,\overline{a_u}^{pr}(\overline{C_{\phi1\phi2}}\,a_u^{sr}-C_{\phi1^2}\tilde{\mu}y_u^{sr})\,LF_{2,2,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+$$
$$\frac{3}{8}\tilde{\mu}(3\,\overline{C_{\phi1\phi2}}\,\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}+$$
$$\overline{a_u}^{pr}(-\overline{C_{\phi1\phi2}}\,\overline{y_u}^{st}a_u^{pt}a_u^{sr}+y_u^{sr}(C_{\phi1\phi2}\overline{a_u}^{st}a_u^{pt}+\tilde{\mu}\,\overline{y_u}^{st}(4C_{\phi1^2}a_u^{pt}+C_{\phi1\phi2}\tilde{\mu}y_u^{pt}))))$$
$$LF_{2,2,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]+3(\tilde{\mu}^3\,\overline{y_d}^{pr}\,\overline{y_u}^{st}y_u^{pt}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde{\mu}y_d^{sr})+$$
$$\overline{a_d}^{pr}a_u^{pt}(-\tilde{\mu}\,\overline{y_u}^{st}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde{\mu}y_d^{sr})+\overline{a_u}^{st}(C_{\phi2^2}a_d^{sr}+C_{\phi1\phi2}\tilde{\mu}y_d^{sr})))$$
$$LF_{2,2,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]+\frac{3}{8}(\overline{C_{\phi1\phi2}}\,\tilde{\mu}^3\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}+$$
$$\tilde{\mu}\,\overline{a_u}^{pr}(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{st}a_u^{pt}a_u^{sr}+y_u^{sr}(-C_{\phi1\phi2}\overline{a_u}^{st}a_u^{pt}+\tilde{\mu}\,\overline{y_u}^{st}(4C_{\phi1^2}a_u^{pt}+3C_{\phi1\phi2}\tilde{\mu}y_u^{pt}))))$$
$$LF_{2,2,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+3\tilde{\mu}^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(C_{\phi1^2}+C_{\phi2^2})$$
$$LF_{2,1,1,1,0}[m_{\tilde{u}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]-$$
$$3\tilde{\mu}^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(C_{\phi1^2}+C_{\phi2^2})\,LF_{3,1,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]-\frac{3}{2}\tilde{\mu}\,\overline{y_u}^{st}a_u^{pt}$$
$$(\overline{C_{\phi1\phi2}}\,\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{sr}+\overline{a_u}^{pr}(\overline{C_{\phi1\phi2}}\,a_u^{sr}+\tilde{\mu}y_u^{sr}(C_{\phi1^2}+C_{\phi2^2})))\,LF_{2,1,1,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$
$$\frac{3}{2}\tilde{\mu}\,\overline{y_u}^{st}a_u^{pt}(\overline{C_{\phi1\phi2}}\,\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{sr}+\overline{a_u}^{pr}(\overline{C_{\phi1\phi2}}\,a_u^{sr}+\tilde{\mu}y_u^{sr}(C_{\phi1^2}+C_{\phi2^2})))$$
$$LF_{3,1,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+\frac{3}{4}(\overline{C_{\phi1\phi2}}\,\tilde{\mu}^3\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}+\tilde{\mu}\,\overline{a_u}^{pr}$$
$$(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{st}a_u^{pt}a_u^{sr}+y_u^{sr}(C_{\phi1\phi2}\overline{a_u}^{st}a_u^{pt}+\tilde{\mu}\,\overline{y_u}^{st}(2a_u^{pt}(C_{\phi1^2}+C_{\phi2^2})+C_{\phi1\phi2}\tilde{\mu}y_u^{pt}))))$$
$$LF_{2,2,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]-3(C_{\phi1^2}+C_{\phi2^2})$$
$$(\tilde{\mu}^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}+\overline{a_d}^{pr}a_u^{pt}(a_d^{sr}\,\overline{a_u}^{st}-\tilde{\mu}^2y_d^{sr}\,\overline{y_u}^{st}))\,LF_{2,1,1,1,0}[m_{\tilde{u}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+$$
$$3(C_{\phi1^2}+C_{\phi2^2})(\tilde{\mu}^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}+\overline{a_d}^{pr}a_u^{pt}(a_d^{sr}\,\overline{a_u}^{st}-\tilde{\mu}^2y_d^{sr}\,\overline{y_u}^{st}))$$
$$LF_{3,1,1,1,-1}[m_{\tilde{u}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]-\frac{3}{2}\tilde{\mu}y_u^{sr}\,\overline{a_u}^{pr}$$
$$(C_{\phi1\phi2}\overline{a_u}^{st}a_u^{pt}+\tilde{\mu}\,\overline{y_u}^{st}(a_u^{pt}(C_{\phi1^2}+C_{\phi2^2})+C_{\phi1\phi2}\tilde{\mu}y_u^{pt}))\,LF_{2,1,1,1,0}[m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+$$
$$\frac{3}{2}\tilde{\mu}y_u^{sr}\,\overline{a_u}^{pr}(C_{\phi1\phi2}\overline{a_u}^{st}a_u^{pt}+\tilde{\mu}\,\overline{y_u}^{st}(a_u^{pt}(C_{\phi1^2}+C_{\phi2^2})+C_{\phi1\phi2}\tilde{\mu}y_u^{pt}))$$
$$LF_{3,1,1,1,-1}[m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}])$$